

AlphaSimR Workshop

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Minneapolis, MN
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Agenda

Self introduction

Class introductions

Class overview

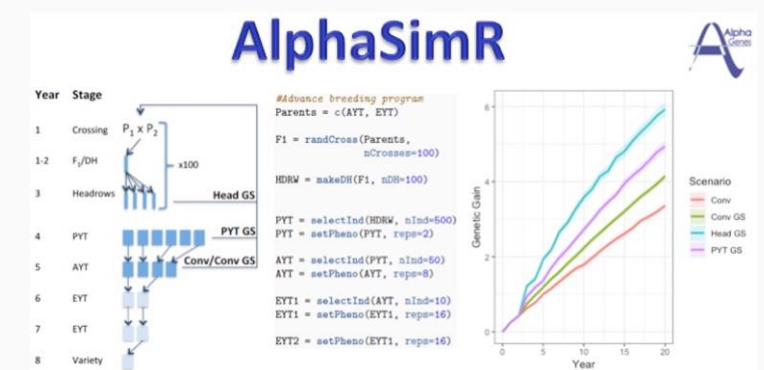
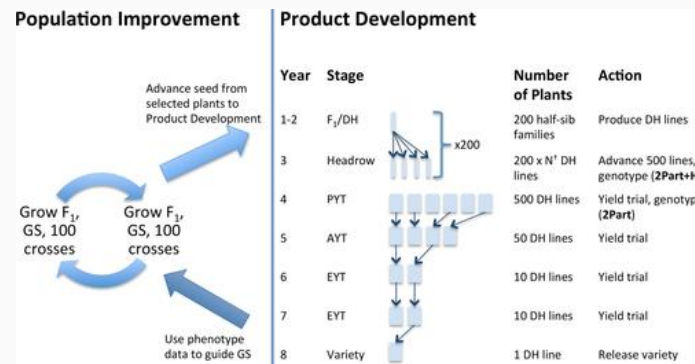
What is stochastic simulation

Introduction to AlphaSimR



Chris Gaynor – My History and Experience

- ♦ Oregon State University
 - ♦ BS Biochemistry
 - ♦ Assistant Wheat Breeder
 - ♦ MS Crop Science
- ♦ Kansas State University
 - ♦ PhD Genetics
- ♦ University of Edinburgh
 - ♦ Post-doc Quant. Gen.
- ♦ Bayer
 - ♦ Sr Data Scientist
 - ♦ Principal Quant. Gen.



Class Introductions

Name

University

Current research project(s)

What do you want from this class?



Class overview

All files are here: https://github.com/gaynorr/AlphaSimR_Class

- There will be a mix of different types of sections
 - R scripts with me demonstrating code and you working on example problems
 - A challenge problem
 - Theory lectures with code demonstration
- Ask questions at anytime
- We will have two 30-minute breaks at 10 and 2:30
- We will have a 1-hour lunch at 12

What is stochastic simulation?

- ♦ There are two types of simulation
 - ♦ Deterministic
 - ♦ Provides a fixed output
 - ♦ Stochastic
 - ♦ Incorporates randomness
- ♦ Deterministic simulations are common
 - ♦ Often used in textbooks
 - ♦ Breeder's equation
 - ♦ Optimization of multistage testing
- ♦ Stochastic simulations can replace deterministic

Breeder's equation (R code)

$$Gain = \frac{ih\sigma_A}{L}$$

- ♦ i : Selection intensity
- ♦ h : Accuracy
- ♦ σ_A : Diversity
- ♦ L : Generation interval

Deterministic

$h^2 = 0.5$

$V_a = 1$

$i = \text{dnorm}(\text{qnorm}(0.9)) / 0.1$ # top 10%

$i * \text{sqrt}(h^2) * \text{sqrt}(V_a)$ # 1.240961

Stochastic

$a = \text{rnorm}(10000, \text{sd}=1)$ # additive genetic values, $V_a = 1$

$e = \text{rnorm}(10000, \text{sd}=1)$ # non-additive values

$p = a + e$ # phenotype, $h^2 = 0.5$

$\text{best} = \text{order}(p, \text{decreasing}=\text{TRUE})[1:1000]$ # top 10%

$\text{mean}(a[\text{best}]) - \text{mean}(a)$ # ~ 1.240961

Why use stochastic simulation?

- ♦ Doesn't require deterministic formulas
 - ♦ Many things of interest lack robust formulas
 - ♦ Genomic prediction accuracy
 - ♦ Long-term response to selection
- ♦ Better handles complexity
 - ♦ Whole breeding program simulations
 - ♦ Can answer many more questions
- ♦ Modern computing makes them feasible

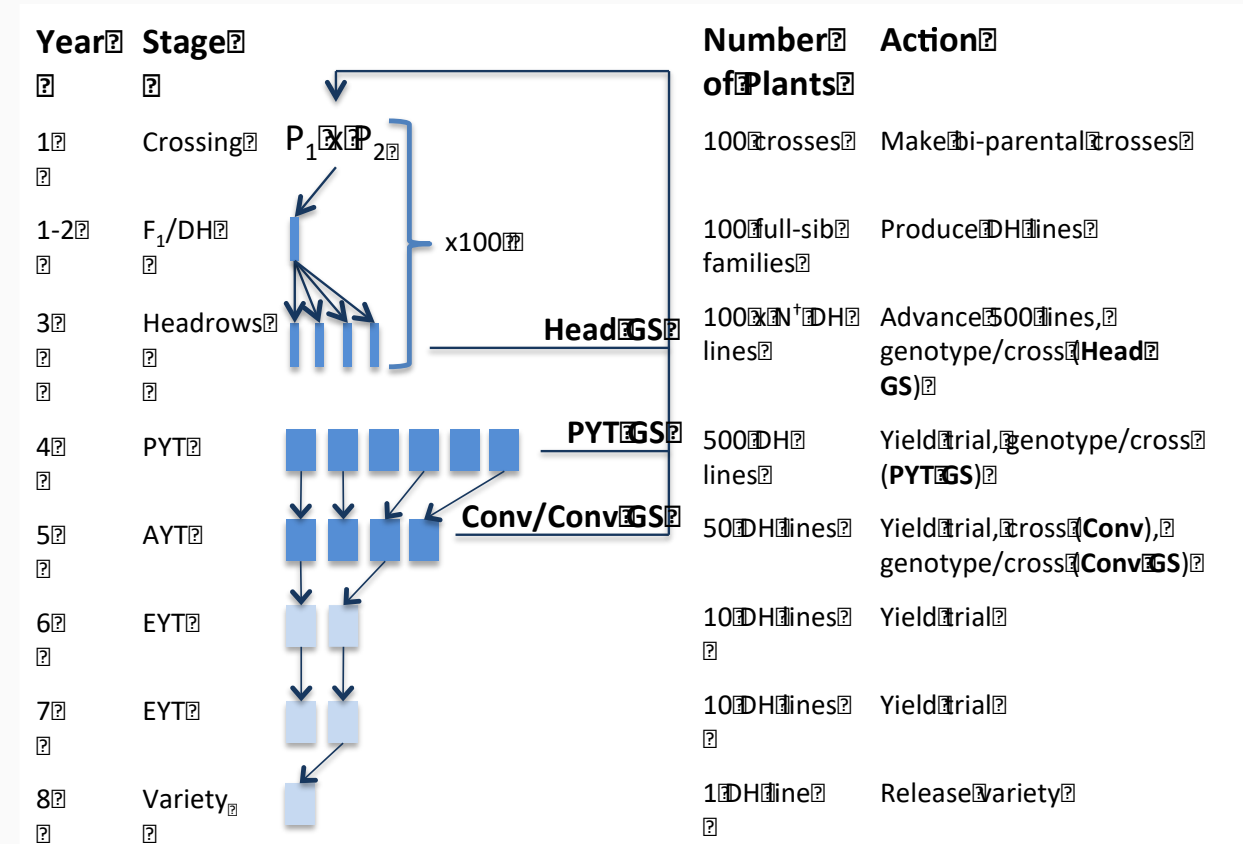
Example: Two-part strategy for using genomic selection

- ♦ Breeding programs have two components
 - ♦ Population improvement
 - ♦ Selection of new parents to improve germplasm
 - ♦ Product development
 - ♦ Development of new varieties or hybrids
- ♦ Population improvement is typically part of the product development pipeline
- ♦ Can we improve genetic gain by separating these processes?
 - ♦ Two-part strategy
 - ♦ Rapid cycle genomic selection for population improvement
 - ♦ Traditional breeding program for product development

Gaynor, R. C., G. Gorjanc, A. R. Bentley, E. S. Ober, P. Howell *et al.*, 2017 A Two-Part Strategy for Using Genomic Selection to Develop Inbred Lines. *Crop Science* 57: 2372–2386.

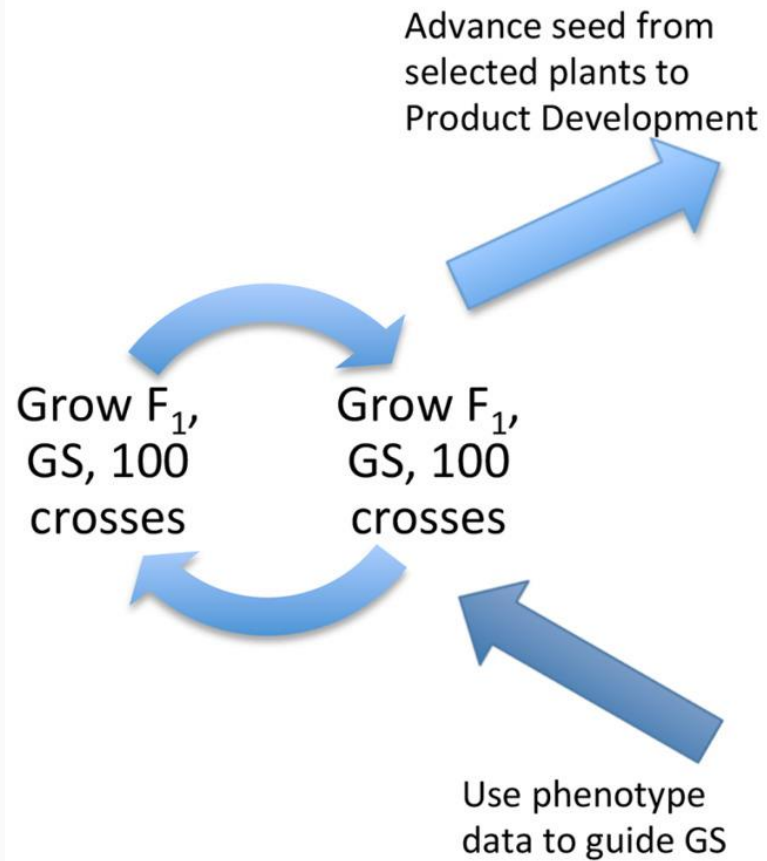
Conventional Strategies

- Roughly based on UK wheat programs
- A purely phenotypic program for control
 - Conv
- Adding genomic selection to the program
 - To increase accuracy
 - Conv GS
 - To reduce cycle time
 - PYT GS and Head GS

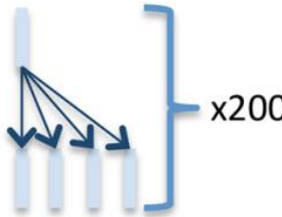
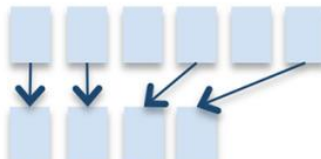
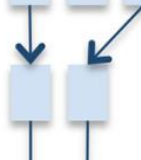


Two-part strategy

Population Improvement



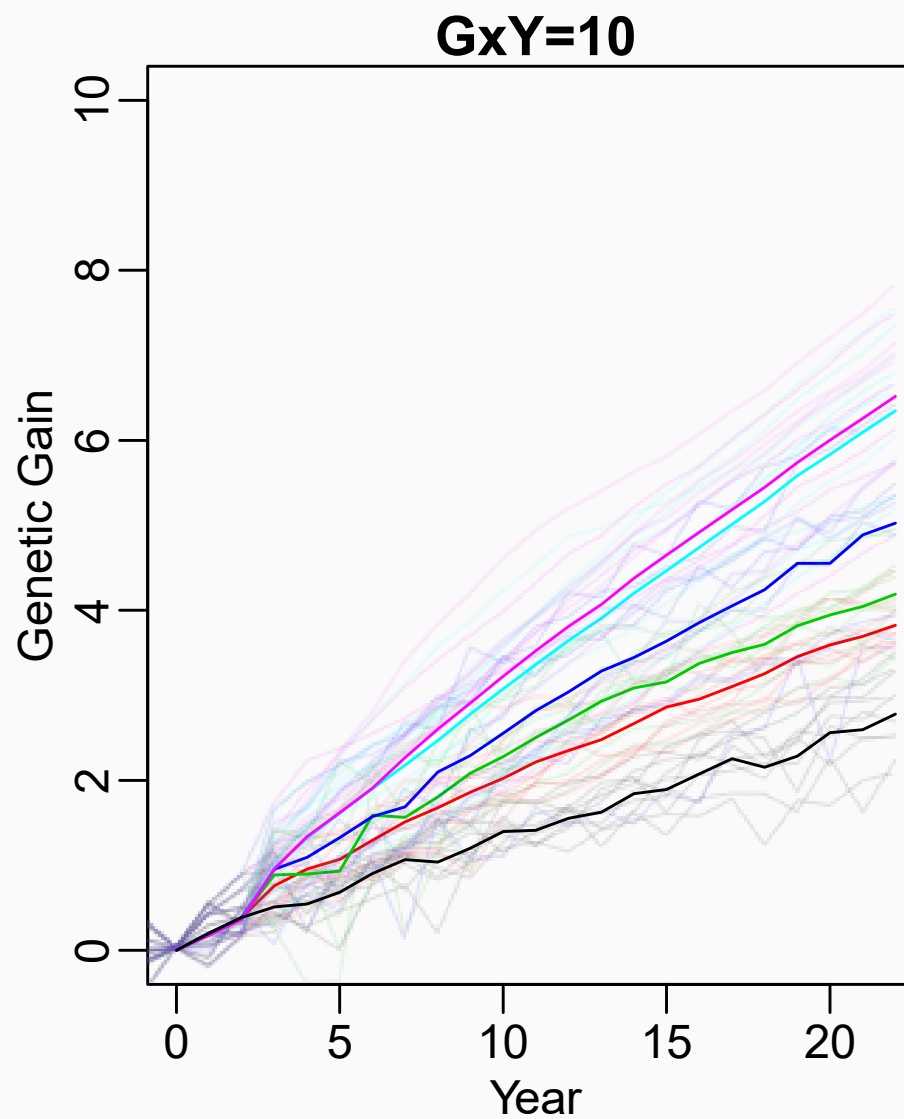
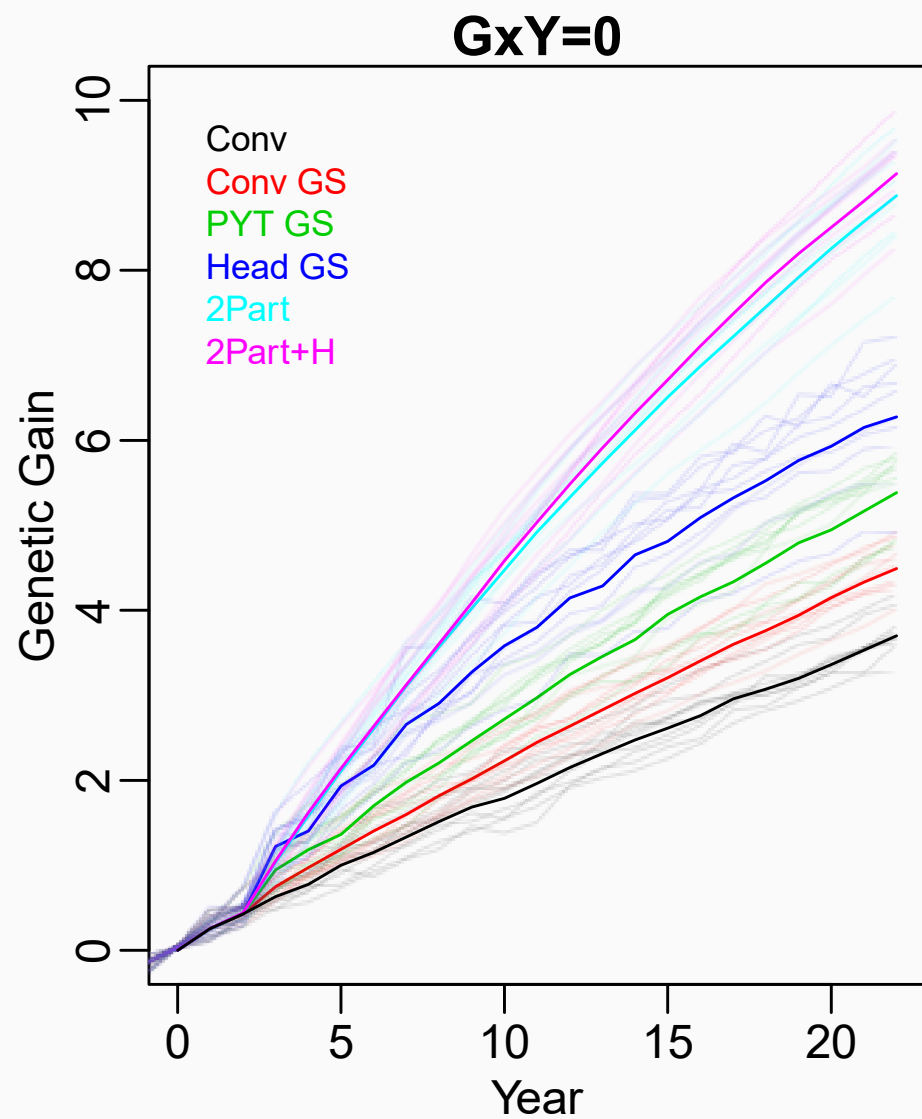
Product Development

Year	Stage		Number of Plants	Action
1-2	F_1 /DH		200 half-sib families	Produce DH lines
3	Headrow		200 $\times$ N ⁺ DH lines	Advance 500 lines, genotype (2Part+H)
4	PYT		500 DH lines	Yield trial, genotype (2Part)
5	AYT		50 DH lines	Yield trial
6	EYT		10 DH lines	Yield trial
7	EYT		10 DH lines	Yield trial
8	Variety		1 DH line	Release variety

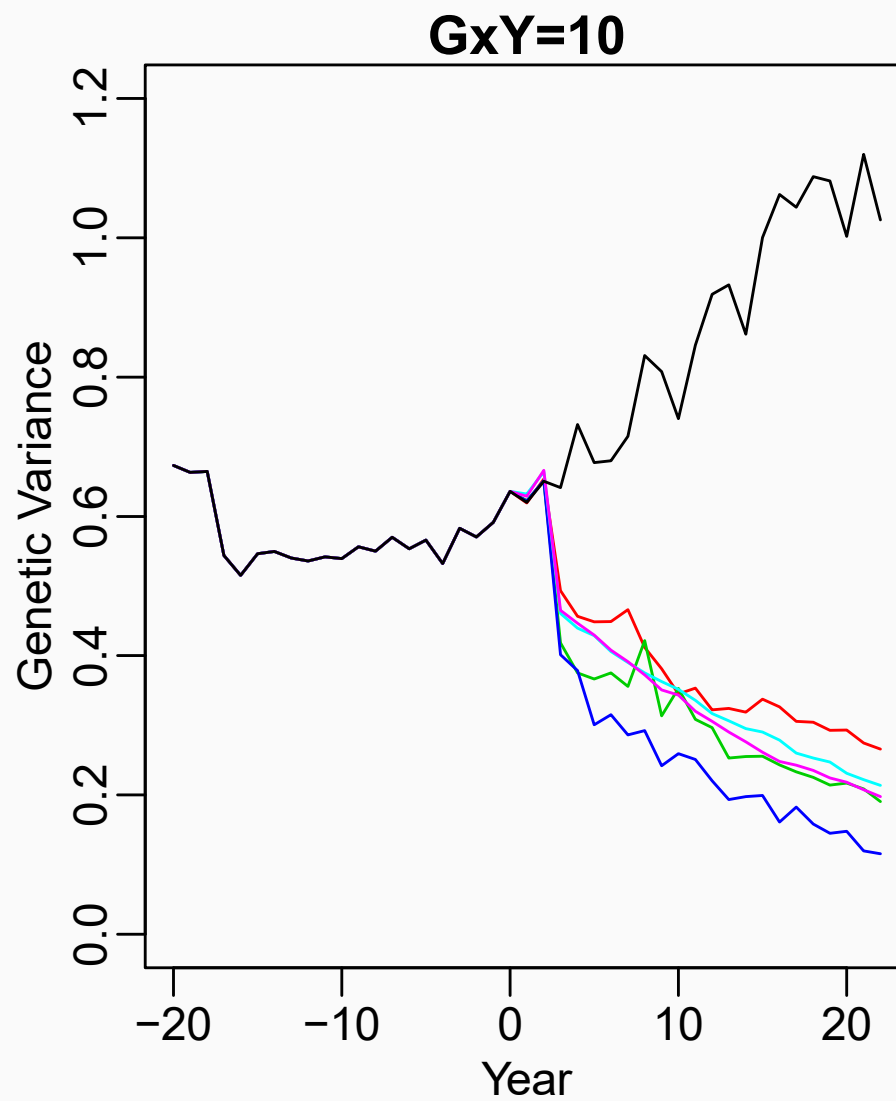
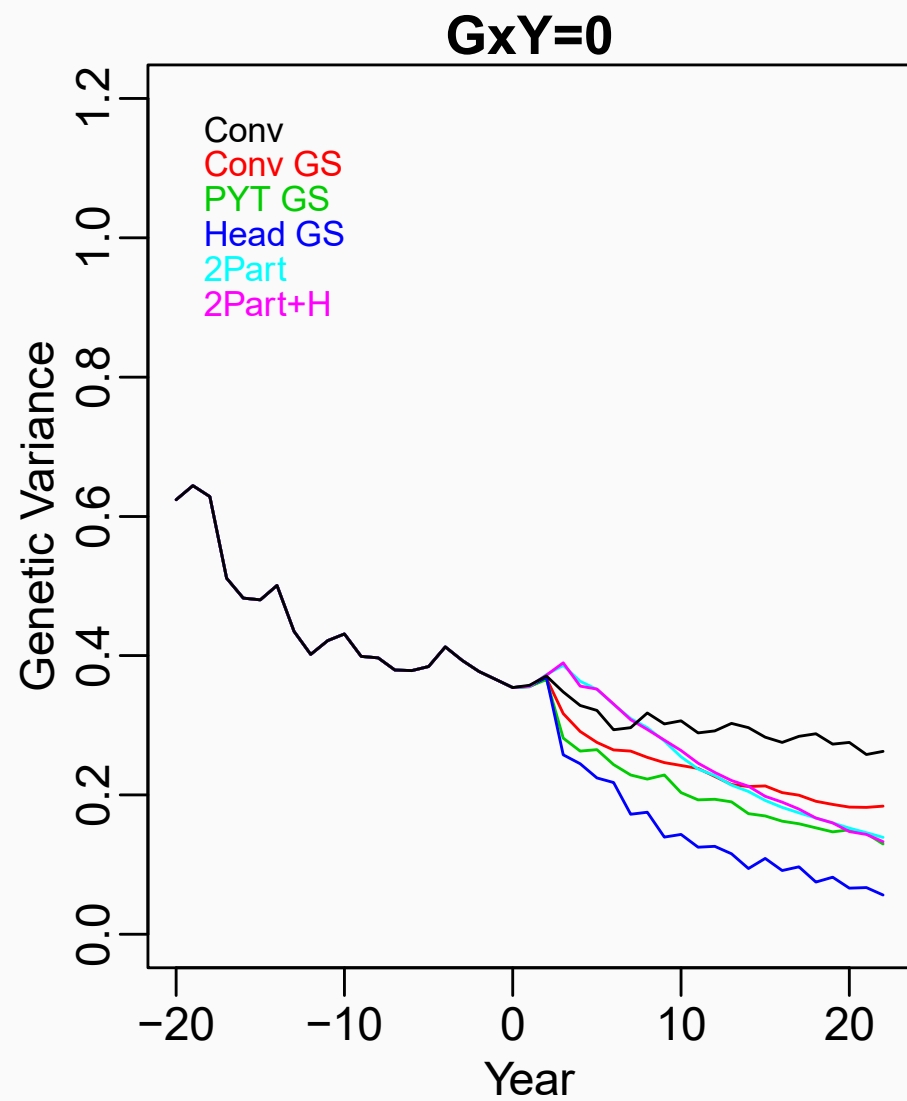
Simulation scheme

Simulation Stage		Key Features
Burn-in	Genome Sequence	100,000 generations of evolution Wheat historical effective population size 21 chromosome pairs 1.43 Morgans per chromosome 8×10^8 base pairs per chromosome 2×10^{-9} mutation rate
	Founder Genotypes	50 inbred founders 21,000 SNP markers 21,000 QTN Normally distributed QTN effects
	Recent Breeding	20 years of modern breeding (-19 to 0) Double haploid lines No genomic selection
Evaluation	Future Breeding	20 years of breeding (1 to 20) Testing alternative breeding programs Equal cost programs Ridge regression for genomic selection

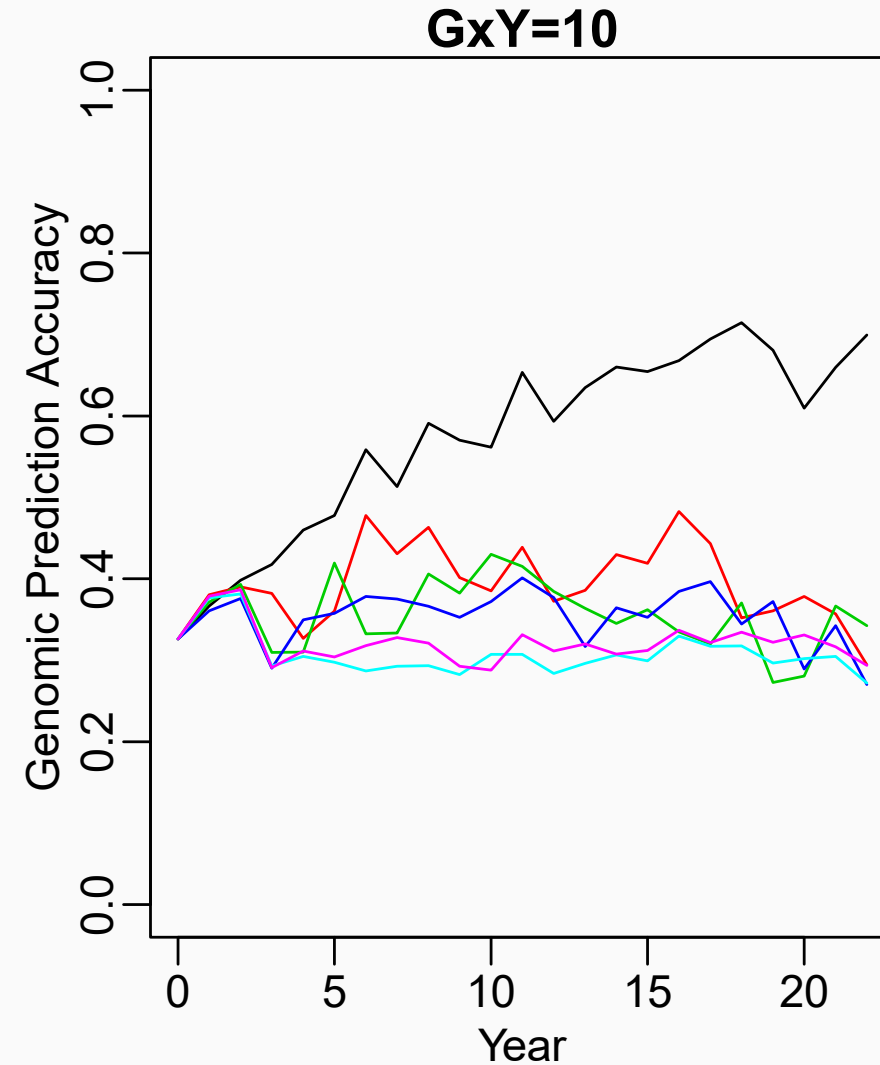
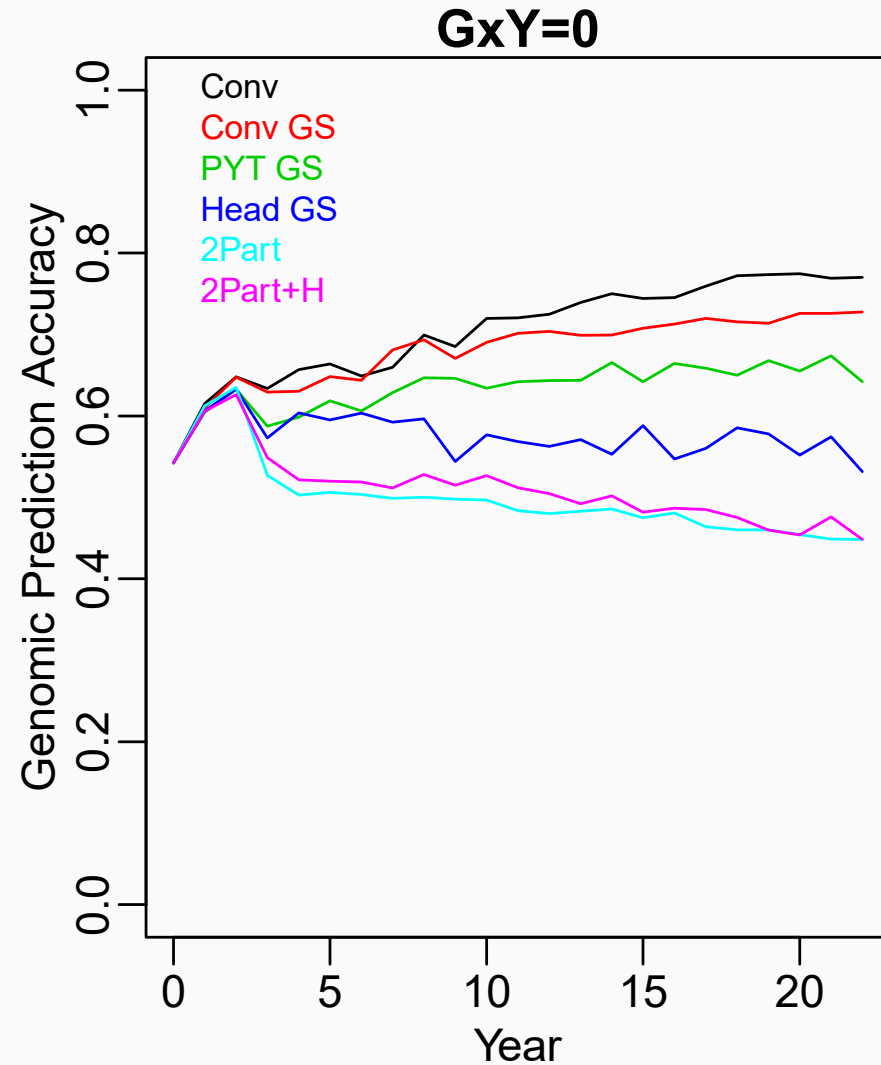
Genetic gain



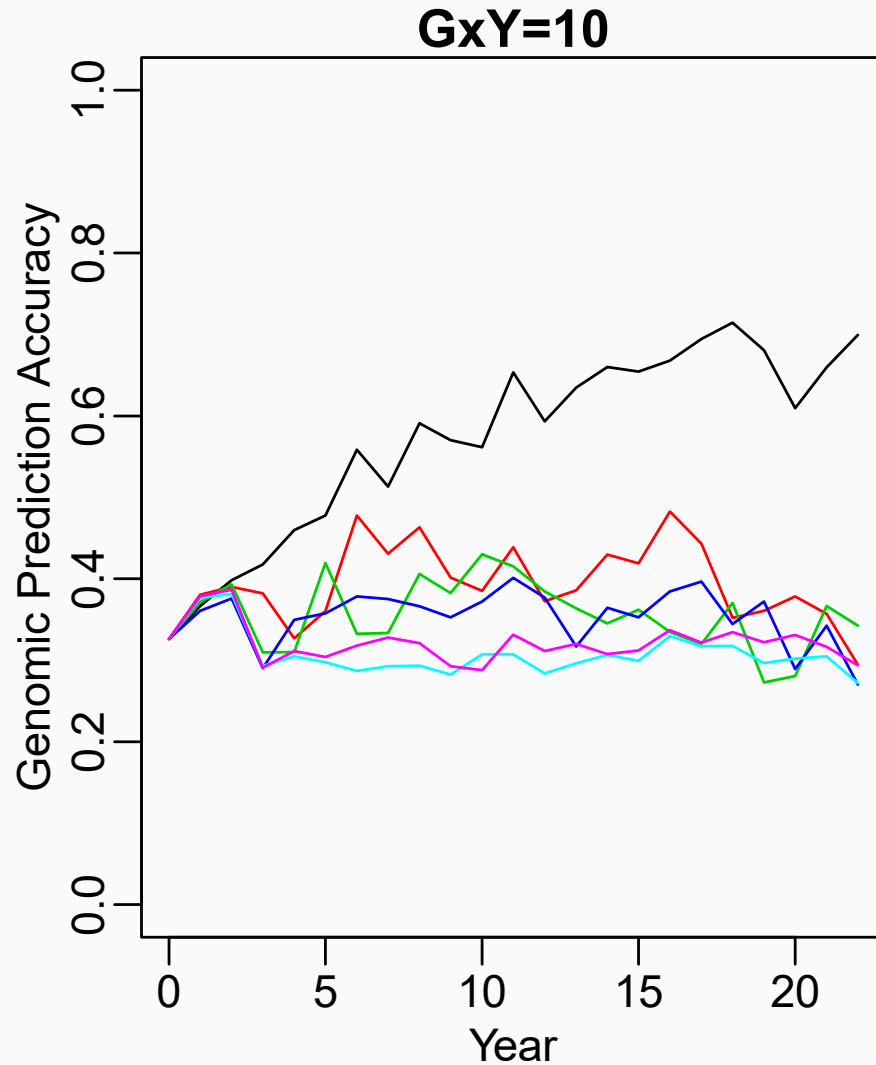
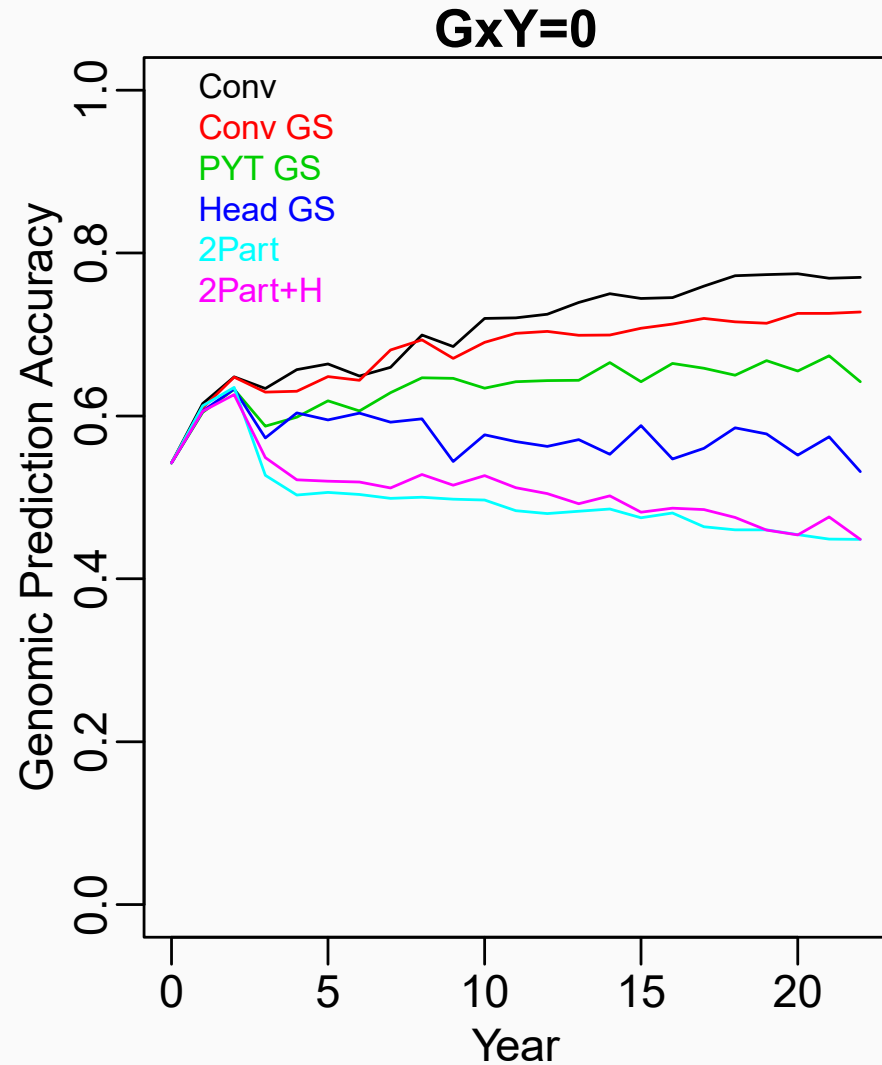
Genetic diversity



Genomic selection accuracy, headrow stage



Genomic selection accuracy, headrow stage



Conclusions

- ♦ Simulation shows massive potential
- ♦ GS accuracy may be too high
 - ♦ No epistasis
 - ♦ Overly persistent accuracy
 - ♦ Bias towards two-part
- ♦ Simulation doesn't consider all risks
 - ♦ Operational risk
 - ♦ Selection on multiple traits

Introduction to AlphaSimR

- ♦ The previous simulation used AlphaSim
- ♦ AlphaSim had several limitations
 - ♦ Steep learning curve
 - ♦ Difficult-to-maintain source code
 - ♦ Not specifically designed for breeding program simulations
- ♦ AlphaSimR was built from scratch to replace it
 - ♦ Incorporates lessons learned from AlphaSimR
 - ♦ Designed specifically for breeding program simulations

Gaynor, R. C., G. Gorjanc, and J. Hickey. 2021. AlphaSimR: an R package for breeding program simulations. G3 11(2), jkaa017.

Simulations in AlphaSim

- Options set in a specification file
- Lots of external scripting for big jobs
- All output written to disk
- Simple trait models
- Poor recombination model

AlphaSim Specification File

```
#####
BOX 1 - PEDIGREE
PedigreeStatus                      ,Internal
NumberGenerationsBurnIn             ,1
NumberGenerationsSelection          ,2
NumberOfIndividualsInEachGeneration ,Variable,50,10,100
#####
BOX 2 - ANIMAL BREEDING OPTIONS
AnimalBreedingOnOff                 ,Off
NumberOfSiresInEachGeneration       ,Constant,0
NumberOfDamsInEachGeneration        ,Constant,0
#####
BOX 3 - PLANT BREEDING OPTIONS
PlantBreedingOnOff                  ,On
PerformMatingsConsideringTheGenderYesNo ,No
NumberOfMaleParentsInEachGeneration ,Constant,0
NumberOfFemaleParentsInEachGeneration ,Constant,0
NumberOfParentsRegardlessTheGenderInEachGen ,Variable,2,0
PerformSelfingInAtLeastOneGenerationYesNo ,Yes
NumberOfSelfedParentsInEachGeneration ,Variable,0,10
NumberOfGenerationsIncludingDoubledHaploids ,2
CreateDoubledHaploidsInTheseGenerations ,1,3
#####
...

```

Simulations in AlphaSimR

- ♦ Simulations written as R scripts
- ♦ Output on demand
- ♦ Expanded trait models
- ♦ Better recombination models
- ♦ Available on CRAN

AlphaSimR Script

```
founderPop = runMacs(nInd=50, nChr=21,
                    segSites=1000, inbred=TRUE)

SP = SimParam$
  new(founderPop)$
  addTraitA(1000)$
  setVarE(H2=0.4)

Parents = newPop(founderPop)
F1 = randCross(Parents, 200)

HDRW = makeDH(F1, 100)
HDRW = setPheno(HDRW, varE=9) #H2=0.1

PYT = selectWithinFam(HDRW, 5)
PYT = setPheno(PYT)

AYT = selectInd(PYT, 100)
AYT = setPheno(AYT, reps=4)

EYT = selectInd(AYT, 10)
EYT = setPheno(EYT, reps=16)

Variety = selectInd(EYT, 1)
```

Overview of AlphaSimR

- ♦ Designed to be intuitive and flexible
 - ♦ R-like objects and interface
 - ♦ Populations as the basic unit
 - ♦ Intuitive function names
- ♦ Designed for high performance
 - ♦ Heavy use of C++
 - ♦ Multithreading with OpenMP
 - ♦ Bitwise storage of genotypes
 - ♦ On demand calculations and data storage

Genetic recombination in AlphaSimR

- ♦ Modeling of crossover interference
 - ♦ Gamma “sprinkling” model
 - ♦ Default parameters match Kosambi function
 - ♦ No “sprinkling”
 - ♦ Can be adjusted by the user
- ♦ Supports autopolyploids
 - ♦ Any level of ploidy
 - ♦ Tunable levels of multivalent pairing
 - ♦ Bivalent pairing
 - ♦ “Cross-type” quadrivalents
 - ♦ Double reduction possible

Traits in AlphaSimR

- ♦ Classic trait models
 - ♦ **ADEG** framework
- ♦ Any number of traits or QTL
 - ♦ Pleotropic, correlated traits
- ♦ User supplies trait mean and variance
 - ♦ Software scales effects to match values

$$GV(x, w) = \mu + A(x) + D(x) + E(x) + G(x, w)$$

- ♦ **A**dditive
- ♦ **D**ominance
 - ♦ Digenic
- ♦ **E**pistasis
 - ♦ Additive-by-additive
- ♦ **G**enotype-by-environment
 - ♦ One environmental covariate

Traits in autopolyploids

- ♦ Genotype dosage scaling used for polyploids
 - ♦ Done for modeling purposes
 - ♦ Not biological accurate
- ♦ Raw genotype dosage goes from 0 to ploidy (x)
- ♦ Additive genotype dosage

$$x_A = \left(x - \frac{ploidy}{2}\right) \left(\frac{2}{ploidy}\right)$$
- ♦ Dominance genotype dosage

$$x_D = x(ploidy - x) \left(\frac{2}{ploidy}\right)^2$$

Table 1: Raw and scaled genotype dosages.

Diploid	Tetraploid	Additive	Dominance
0	0	-1	0
	1	-1/2	3/4
1	2	0	1
	3	1/2	3/4
2	4	1	0

Additive and dominance effects

- ♦ Additive effects
 - ♦ Two possible distributions
 - ♦ Gaussian or gamma with random sign
 - ♦ Mean is zero, variance scaled automatically

$$A(x) = \sum a x_A$$

- ♦ (Digenic) Dominance effects
 - ♦ User specifies dominance degrees (delta)
 - ♦ Mean and variance of gaussian distribution

$$D(x) = \sum d x_D \qquad d = \delta |a|$$

Table 2: Example genetic values ($a = 1$, $d = 1$).

Diploid Dosage	Tetraploid Dosage	Genetic Value
0	0	-1
	1	1/4
1	2	1
	3	5/4
2	4	1

Epistatic and genotype-by-environment interactions

- ♦ Limited modeling of epistasis
 - ♦ Additive-by-additive epistasis
 - ♦ Between discrete pairs of loci
 - ♦ Half as many interactions as loci
- ♦ User sets “relative” variance
 - ♦ Ratio between epistatic and additive variance
 - ♦ If allele frequency is 0.5
- ♦ Simple genotype-by-environment model
 - ♦ Additive effects by a single covariate
- ♦ User sets variances for base population
 - ♦ Environmental
 - ♦ Genotype-by-environment
- ♦ Covariate passed to functions as a p-value
 - ♦ User can set p-value
 - ♦ Otherwise sampled at random
- ♦ Typically used for genotype-by-year interactions

Variance Components

- ♦ Full decomposition of variance components
 - ♦ Additive, dominance, and epistatic
 - ♦ Genic and genetic variations
 - ♦ See Bulmer, 1976
- ♦ Calculations for any population structures
 - ♦ Doesn't require Hardy-Weinberg assumption
 - ♦ Accounts of LD
 - ♦ Handles any ploidy level

Genomic selection

- ♦ Several built-in solvers
 - ♦ RR-BLUP based models
 - ♦ Selected for speed and ease-of-use
 - ♦ Variations available for hybrid crops
- ♦ Can use other programs
 - ♦ Other R packages
 - ♦ External software

Example: optimal number of parents and crosses

- ♦ Simulation for IITA east African cassava program
- ♦ Varied number of parents and crosses
 - ♦ Three parent composition scenarios
 - ♦ PYT (all from first yield trial)
 - ♦ AYT (all from second yield trial)
 - ♦ Mix (50/50 from first two yield trials)
 - ♦ Crosses made at random
 - ♦ Size of program remained constant
 - ♦ 1,632 total progeny
- ♦ Produce some of my favorite figures

Figure 1: Number of crosses and parents

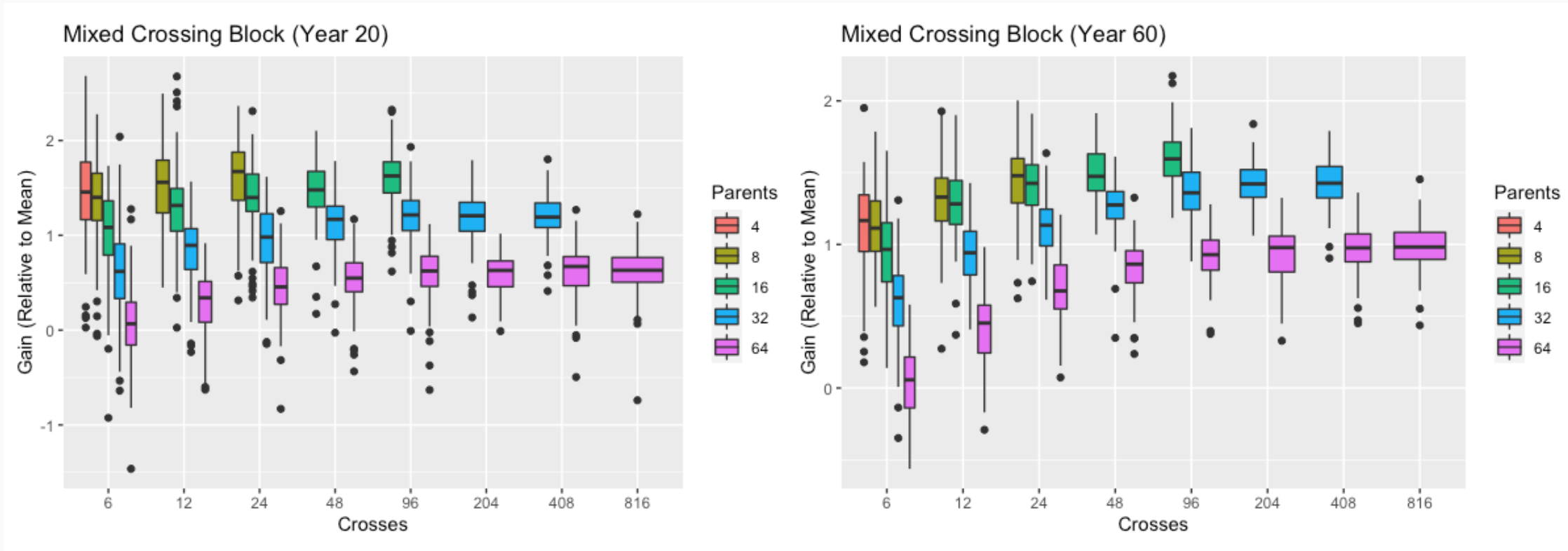
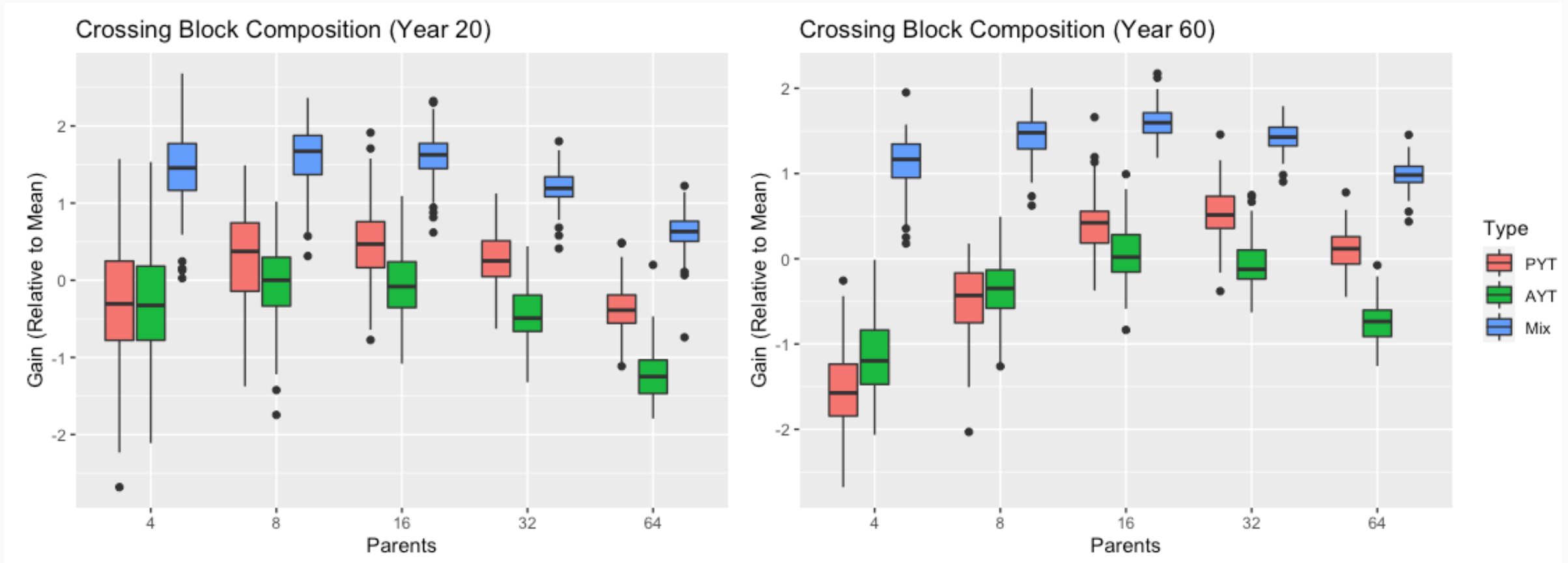


Figure 2: Parent composition strategy



What do these plots tell us

- ♦ A small number of parents is good for short-term gain
 - ♦ Due to high selection intensity
 - ♦ Exposes you to drift
 - ♦ Hurts long-term gain
 - ♦ Adds risk (not fully captured by simulation)
- ♦ More crosses is better the more progeny per cross
 - ♦ Assuming crosses are random
 - ♦ Assuming similar cost
- ♦ It is good to mix cohorts when crossing
 - ♦ Effectively increases breeding program size
 - ♦ Buffers against genotype-by-year interactions

Considerations and recommendations

- ♦ Simulations need guard rails
 - ♦ Be mindful of biological and operational limits
 - ♦ Testing outside these limits is sometimes helpful
- ♦ Avoid over complicating simulations
 - ♦ Don't test too many variables at once
 - ♦ You can't fully capture reality
 - ♦ Focus on the most critical questions

Considerations and recommendations (cont'd)

- ♦ Interesting quantitative genetic properties are observable
 - ♦ Heterosis and inbreeding depression
 - ♦ Using directional dominance
 - ♦ Progressive heterosis in autopolyploids
 - ♦ Multiallelic loci are not required
 - ♦ Hybrid depression
 - ♦ Using additive-by-additive epistasis
- ♦ I don't observe epistatic decay
 - ♦ Inbred progeny average less than midparent
 - ♦ This occurs frequently in reality with elite parents
 - ♦ May result in upwardly biased GS accuracy

Considerations and recommendations (cont'd)

- ♦ I recommend treating simulation as an iterative process
 - ♦ Empirical data should guide simulations
 - ♦ Simulations should guide collection of empirical data
- ♦ Use simulations to make and test plans
 - ♦ Rerun when details change
 - ♦ Simulation can guide mitigation steps

Links to additional resources

- ♦ <https://doi.org/10.1002/csc2.21312>
 - ♦ Plant breeding simulations with AlphaSimR
- ♦ <https://tinyurl.com/jyy8bj8u>
 - ♦ Massive online open course for AlphaSimR
- ♦ <https://github.com/gaynorr/AlphaSimR>
 - ♦ Use “Discussions” to post questions
- ♦ https://github.com/gaynorr/AlphaSimR_Examples
 - ♦ Example scripts of variable quality
- ♦ <https://excellenceinbreeding.org/toolbox/collection/244>
 - ♦ Excellence in breeding simulation library

Acknowledgments

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